

The Cowan Atomic-Structure Codes

What RCN, RCN2, RCG, RCE Actually Compute, from First Principles

A working explainer for the Modern Atomic Linelists (`mal`) project

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Abstract

This document explains, from first principles, what the four Cowan programs do and how they chain together to produce atomic energy levels and oscillator strengths (*gf*-values) — the data that go into stellar line lists. It maps the physics onto the *actual* programs, input decks, and output files in this repository (`cowan_nist/extracted/`). The goal is a shared, durable understanding of the engine we are modernizing, with particular attention to the least-squares fitting stage (RCE), which is the heart of the project. The canonical reference throughout is R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981), cited as **TASS**.

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1 The big picture: why these four programs exist

A stellar line list needs, for each transition, the lower and upper energy levels, the wavelength, and the oscillator strength *gf*. The wavelength is just the level difference; the levels and the *gf* are the hard part. Computing them *ab initio* for complex atoms (open *d*- and *f*-shells, the iron group) is not reliably accurate. Cowan’s insight, embodied in these codes, is a *semi-empirical* scheme: compute the atomic structure approximately from theory, then *adjust a small number of physically meaningful radial parameters until the predicted levels match the experimentally measured ones*. The wavefunctions (eigenvectors) that result are far better than the purely theoretical ones, and the *gf*-values computed from them are correspondingly better.

The pipeline in one breath: RCN solves for the radial wavefunctions and the *radial integrals* ($F^k, G^k, \zeta_{nl}, \dots$) of each electron configuration; RCN2 forms the configuration-interaction and

electric-multipole (transition) radial integrals between configurations; RCG builds the angular part, assembles and *diagonalizes* the energy (Hamiltonian) matrix to get energy levels and eigenvectors, and from the eigenvectors computes *gf*-values; RCE re-runs the diagonalization inside a *least-squares loop*, varying the radial parameters until the computed levels best fit the observed levels.

The division of labor reflects a deep structural fact about atomic Hamiltonians: the energy of a level separates into **radial** factors (integrals over the one-electron radial wavefunctions — these are the F^k , G^k , ζ_{nl} , R^k) multiplied by **angular** factors (pure angular-momentum algebra, the same for any atom with a given configuration). RCN/RCN2 compute the radial side; RCG computes the angular side and combines them; RCE treats the radial numbers as adjustable.

2 The physics each stage implements

2.1 The central-field starting point and rcn

We seek the eigenstates of the many-electron Hamiltonian — the non-relativistic electronic Hamiltonian plus the leading (spin-orbit) relativistic correction,

$$H = \underbrace{\sum_i \left(-\frac{1}{2} \nabla_i^2 - \frac{Z}{r_i} \right) + \sum_{i < j} \frac{1}{r_{ij}}}_{\text{non-relativistic}} + \underbrace{\sum_i \xi(r_i) (\mathbf{l}_i \cdot \mathbf{s}_i)}_{\text{spin-orbit}}, \quad (1)$$

in atomic units. The *central-field approximation* writes each electron’s wavefunction as a product of a radial part $P_{n\ell}(r)/r$ and angular/spin parts. RCN solves the resulting one-electron radial equations self-consistently for a *single configuration* (e.g. $3d^7$, or $3d^6 4p$), using one of several approximations to Hartree–Fock; in practice the Hartree-plus-statistical-exchange (HX) method or center-of-gravity HF is used, optionally with relativistic corrections (the “HFR” / “HXR” options — important for the iron group and heavier).

A converged set of these single-electron radial functions is shown in Fig. 1: $P_{n\ell}(r) = r R_{n\ell}(r)$ is the quantity the code actually carries on its radial mesh, with $\int_0^\infty P_{n\ell}^2 dr = 1$. The number of radial nodes ($n - \ell - 1$) and the outward march of the orbitals with n are exactly what determine the sizes of the radial integrals below.

From the converged radial wavefunctions $P_{n\ell}(r)$, RCN computes the **radial integrals** that parametrize the energy:

$$F^k(n\ell, n'\ell') = \int \int P_{n\ell}^2(r_1) \frac{r_{<}^k}{r_{>}^{k+1}} P_{n'\ell'}^2(r_2) dr_1 dr_2 \quad (\text{direct Slater integral}), \quad (2)$$

$$G^k(n\ell, n'\ell') = \int \int P_{n\ell}(r_1) P_{n'\ell'}(r_1) \frac{r_{<}^k}{r_{>}^{k+1}} P_{n\ell}(r_2) P_{n'\ell'}(r_2) dr_1 dr_2 \quad (\text{exchange Slater integral}), \quad (3)$$

$$\zeta_{n\ell} = \frac{\alpha^2}{2} \int P_{n\ell}^2(r) \frac{1}{r} \frac{dV}{dr} dr \quad (\text{spin-orbit parameter}), \quad (4)$$

plus E_{av} , the configuration’s center-of-gravity energy. The $r_{<}^k/r_{>}^{k+1}$ factor is the multipole expansion of the electron–electron repulsion $1/r_{12}$; Fig. 2 shows this kernel and the integrand it produces. These few numbers per configuration are the *radial parameters*. Everything semi-empirical happens by treating them as adjustable later (§2.4).

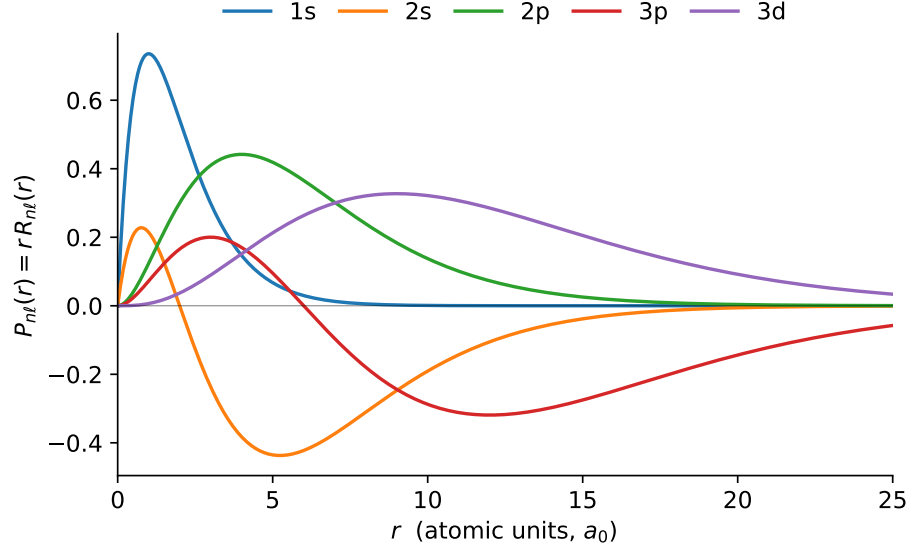


Figure 1: Reduced radial wavefunctions $P_{nl}(r) = r R_{nl}(r)$. (*Illustrative: exact hydrogenic functions, $Z = 1$; a real RCN run produces the self-consistent-field analogues for the actual ion.*) Note the nodal structure — $1s, 2p, 3d$ are nodeless, $2s, 3p$ have one node — and that higher- n orbitals peak at larger radius.

2.2 Configuration interaction and transition integrals: rcn2

Real atomic states are not pure single configurations; nearby configurations of the same parity and J *mix* (configuration interaction, CI). RCN2 takes the radial wavefunctions from RCN for *several* configurations and computes:

- the **CI Slater integrals** R^k (the r_{ij}^{-1} matrix elements *between* different configurations), which set the strength of the mixing (it also computes configuration overlaps and the off-diagonal spin-orbit integrals);
- the **electric-multipole radial integrals** $\langle n\ell | r | n'\ell' \rangle$ (dipole, and higher multipoles), the radial part of the transition moment — the seed of every *gf*-value.

2.3 Angular algebra, diagonalization, and where *gf* comes from: rcg

RCG supplies the *angular* half. Using angular-momentum (Racah/Wigner) algebra and tabulated coefficients of fractional parentage (cfp), it builds the Hamiltonian matrix in a chosen coupling basis. Schematically, every matrix element of (1) is

$$\langle \gamma' | H | \gamma \rangle = \sum_{\text{integrals}} \underbrace{c_{\gamma'\gamma}^{(\cdot)}}_{\text{angular coeff. (RCG)}} \times \underbrace{\{E_{av}, F^k, G^k, \zeta_{nl}, R^k\}}_{\text{radial integrals (RCN/RCN2)}}. \quad (5)$$

RCG then **diagonalizes** this matrix for each J . The eigenvalues are the predicted **energy levels**; the eigenvectors give the **composition** of each level (its percentage breakdown over basis states — e.g. “72% $^5D + 18\% ^3F + \dots$ ”).

The oscillator strength follows from the eigenvectors. The line strength between levels βJ and $\beta' J'$ is

$$S = \left| \langle \beta J \| \mathbf{P}^{(1)} \| \beta' J' \rangle \right|^2, \quad gf = \frac{8\pi^2 m_e c}{3h e^2} \sigma S \propto \sigma S, \quad (6)$$

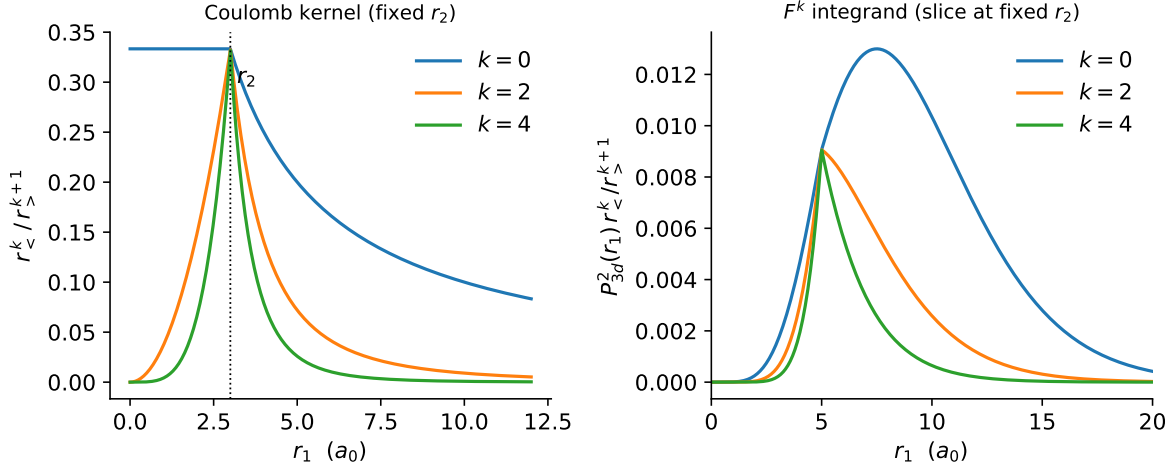


Figure 2: *Left*: the Coulomb multipole kernel $r_{<}^k/r_{>}^{k+1}$ versus r_1 at fixed r_2 — continuous but with a cusp at $r_1 = r_2$, and increasingly short-ranged (peaked near $r_1 = r_2$) for higher k . *Right*: the resulting F^k integrand $P_{3d}^2(r_1) r_{<}^k/r_{>}^{k+1}$ (slice at fixed r_2): the higher- k multipoles weight the region where the two electrons are close, so $F^0 > F^2 > F^4$. (*Hydrogenic 3d, illustrative.*)

where σ is the transition wavenumber and the reduced matrix element of the dipole operator $\mathbf{P}^{(1)}$ is a sum over the RCN2 transition radial integrals weighted by *both* levels' eigenvector components. Here gf is the *weighted* oscillator strength (g already folded in — there is no division by g ; $f = gf/g$ with g the statistical weight of the lower level). In practice RCG evaluates $gf = 3.0375 \times 10^{-6} \sigma S$ with σ in cm^{-1} and S in atomic units ($e^2 a_0^2$). This is the crucial coupling that drives the whole project: **many lines' gf -values depend on the same few radial parameters through the shared eigenvectors.** A small change in the fit redistributes oscillator strength among coupled lines.

2.4 The semi-empirical least-squares fit: rce

This is the stage we are modernizing, so we treat it carefully. RCG's predicted levels are only as good as the theoretical radial integrals — and those are systematically off: Hartree–Fock *overestimates* the Slater integrals by ~ 10 – 20% (so the computed term splittings come out too large; the customary remedy is to scale them down by ~ 0.85). RCE fixes this empirically. Its job (TASS §16-3 to 16-5):

1. Treat the radial parameters $\mathbf{x} = \{E_{\text{av}}, F^k, G^k, \zeta_{nl}, R^k, \dots\}$ as *free*. The angular coefficients in (5) are fixed (precomputed by RCG and passed in the binary file **TAPE2E**).
2. For trial $\mathbf{x}$, build and diagonalize the energy matrix for each J to get computed levels $E_i^{\text{calc}}(\mathbf{x})$.
3. Minimize the least-squares residual against the *experimentally observed* levels E_i^{obs} (from NIST/lab),

$$\chi^2(\mathbf{x}) = \sum_{i \in \text{included}} \left[E_i^{\text{calc}}(\mathbf{x}) - E_i^{\text{obs}} \right]^2, \quad (7)$$

iterating until the parameters stop changing (by < 0.03 in their units between cycles) or a cycle cap is hit. In standard RCE each level is simply *included* or *excluded* (no per-level weights); parameters can also be held fixed, or tied in fixed ratios (a group forced to scale together). Adding per-level or per-line weighting is one of the modernizations we plan, not existing behavior.

4. Feed the *fitted* $\mathbf{x}$ back through the diagonalization to get the *improved* eigenvectors — and hence the improved gf -values (via RCG).

The effect of the fit is sketched in Fig. 3: the raw *ab initio* term pattern is too spread out, and adjusting the radial parameters slides the computed levels onto the observed ones — simultaneously improving every eigenvector, and therefore every gf .

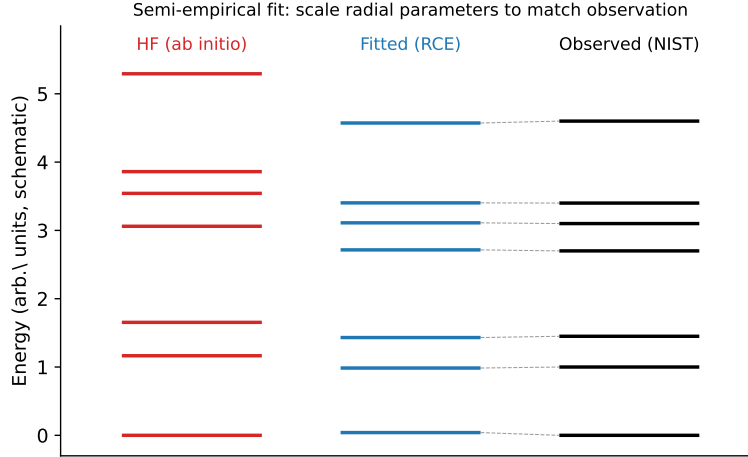


Figure 3: Schematic of the semi-empirical fit. Raw Hartree–Fock levels (left) are systematically too widely split because the Slater integrals are $\sim 15\%$ too large; the RCE least-squares adjustment of the radial parameters (middle) brings the computed levels into agreement with the observed NIST levels (right; dashed guides connect matched levels). (*Schematic, illustrative pattern.*)

The fit in (7) is ill-conditioned: parameters are correlated, the problem is non-linear, and which configurations to include and which parameters to free/fix/tie is a matter of hand-tuned judgment with “no strict methodology.” This is the only non-automated stage, takes hours to weeks per ion, and has scarcely changed in 50 years. It is exactly where modern techniques — regularization, Bayesian inference with priors, MCMC for *per-line gf uncertainties*, global optimization, and (our novel idea) folding the observed *stellar spectrum* into the objective alongside the energy levels — can make a genuine advance. See the project plan.

3 Mapping to the actual code and files

Program	Source (for/)	Computes
RCN	RCN36K.F	Single-config. radial wavefns $P_{n\ell}(r)$; radial integrals $E_{av}, F^k, G^k, \zeta_{n\ell}$
RCN2	RCN2K.F	CI integrals R^k (+ overlaps, off-diagonal spin-orbit); multipole transition integrals $\langle r \rangle$
RCG	rcg11k.f	Angular coeffs (Racah/cfp); builds & diagonalizes $H \rightarrow$ levels, eigenvectors, gf
RCE	RCE20K.F	Least-squares fit of radial parameters to observed levels; improved eigenvectors $\rightarrow gf$

Table 1: The four programs and what they compute. Built by `build/build.sh`.

Data flow between programs. Each program writes the input for the next, mostly through fixed-name files (a legacy of the punched-card/tape era):

- A freshly compiled RCG must first generate, once, the binary “cfp decks” it needs: FOR072 holds the single-subshell quantum numbers and coefficients of fractional parentage; FOR073 and FOR074 hold derived matrix-element and angular-coefficient tables built from them. This is the `build/make_cfp.sh` step (it feeds the shipped `ING11.CFP` deck through RCG).
- In a calculation, RCG writes, among others, `TAPE2E` (binary coefficient matrices for RCE) and `OUTGINE` (a starter for RCE’s text input).
- RCE reads `TAPE2E` plus a text deck `INE20` (control info + the *observed* levels — the only thing not produced upstream) and writes `LEVELS1/2/3` (fitted levels and eigenvectors) and `PARVALS` (fitted parameter values, for restarting).

An actual input deck. The example `work/IN36` is an RCN input for Sn^{7+} (`IN36.ai` is B^{3+} , i.e. B IV , with a long Rydberg series). The first line is a fixed-column “control card” (mesh, method flags, convergence tolerances, relativistic/correlation switches); each subsequent line names one configuration. For Sn^{7+} :

```
200-90 0 2 01. 4.0 5.E-08 1.E-11-2 00090 0 1.0 0.65 0.0 1.00 -6
50 8Sn7+ 4d7 4p6 4D7
50 8Sn7+ 4d65p 4p6 4D6 5P
-1
```

Here the two configurations $4d^7$ and $4d^65p$ (opposite parity — the source and upper levels of E1 transitions) are listed, terminated by `-1`. The columns of this card are documented in `cowan.nist/extracted/RCN.DOC`. we will annotate them in detail when we drive a real run.

4 What “running it” looks like end to end

For a single ion and one parity pair, the canonical chain is

$$\text{RCN} \rightarrow \text{RCN2} \rightarrow \text{RCG} \rightarrow \text{RCE} \rightarrow \text{RCG}$$

(the final RCG re-run applies the fitted parameters to emit final levels and gf). The seven coupling schemes RCG/RCE support (LS, JJ, intermediate, ...) only change the basis, not the physics; final eigenvectors are reported in LS or JJ for interpretation.

Next steps in the project.

1. Drive the shipped example (`IN36`) through the full chain and read each output file together, annotating the real column formats — and replace the illustrative figures here with the actual computed $P_{nl}(r)$ and level fit.
2. Reproduce a known Kurucz Fe II gf via the standard chain (validate we replicate the legacy result).
3. Replace the RCE fit (7) with a modern regularized / Bayesian fit, attach per-line gf uncertainties, and (Tier 2) fold in the solar spectrum.

References

- R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981). [TASS]

- Program documentation shipped with the code: `RCN_DOC.txt`, `RCG_DOC.txt`, `RCE_DOC.txt` (in `cowan_nist/extracted/`).
- A. Kramida, “Cowan Code: 50 Years of Growing Impact on Atomic Physics,” *Atoms* **7**, 64 (2019).